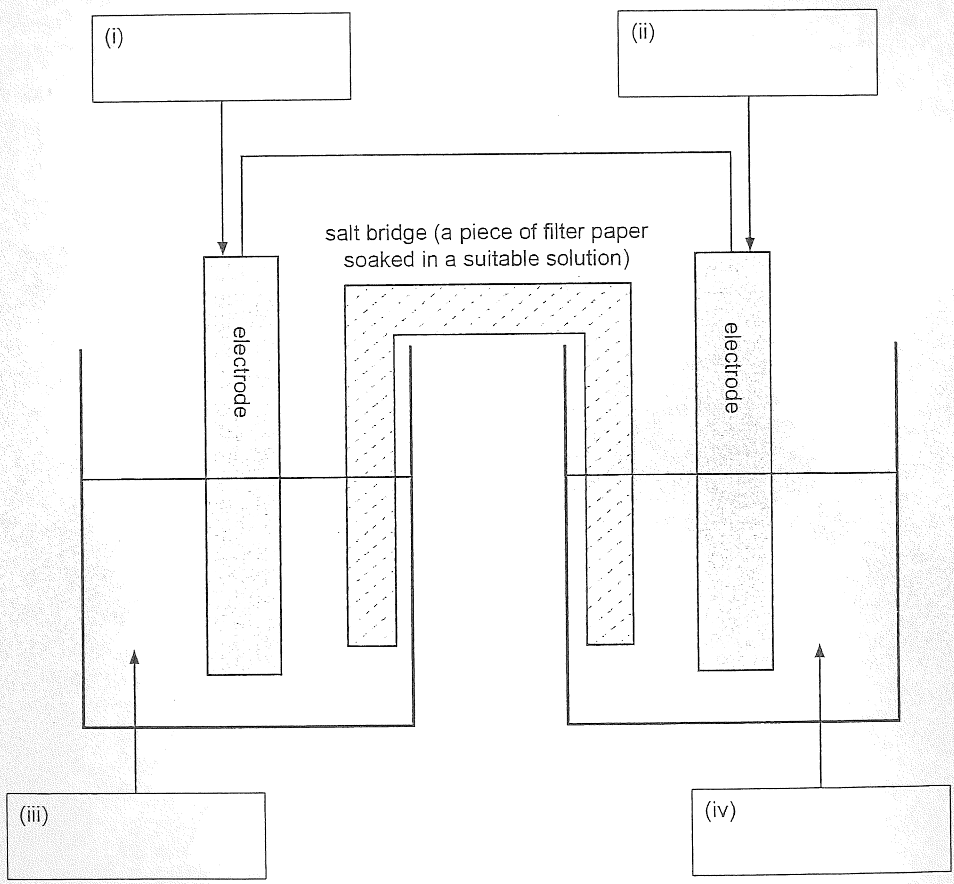
10. **[16 marks]** *4.2.6, 4.2.7, 4.2.8, 4.2.9 (2020:36)*

A student was asked to build a functioning galvanic cell, having been provided with all of the required hardware plus the following substances:

- a piece of magnesium measuring 1 mm by 2 cm by 6 cm
- a piece of copper measuring 1 mm by 2 cm by 6 cm
- a 6 cm long graphite (carbon) rod with a diameter of 1 cm
- 1.0 mol L^{-1} sodium carbonate solution
- 1.0 mol L^{-1} magnesium sulfate solution
- 1.0 mol L^{-1} copper(II) sulfate solution.

There was no requirement for the student to use all of these substances.

(a) A partially-labelled diagram of the galvanic cell built by the student is shown below. What substances should the student have used in the parts labelled (i) to (iv) to build a functioning galvanic cell? Write the names of these substances in the boxes provided. [4]



(b) Add arrows to the diagram in part (a) to show the direction of movement of electrons through the external circuit. [1]

(c) Write the half-equations for the reactions occurring at the anode and the cathode in the student's galvanic cell. [4]

Anode half-equation ______________________________

Cathode half-equation ______________________________

(d) Calculate the electrical potential difference of the student's galvanic cell. Assume standard conditions. Include appropriate units in your answer. [2]

(e) Galvanic cells, such as the one shown in the diagram, need a salt bridge.

(i) State why galvanic cells need a salt bridge. [1]

(ii) Describe, with reference to ion movement, how the salt bridge in a galvanic cell works. Also state why ion movement occurs as you have described. [4]

4. **[12 marks]** *3.2.11, 3.2.10 (2021:35)*

Smoking is hazardous to a person's health and one option to help quit smoking is the use of nicotine patches. These patches, when placed on the skin, release small amounts of nicotine with the aim of reducing cigarette craving.

The nicotine content of these patches can be determined by titration. The titrating solution is prepared by mixing perchloric acid ($HC\ell O_4$) with glacial acetic acid, resulting in the following equilibrium:

$$HC\ell O_4 + CH_3COOH \rightleftharpoons CH_3COOH_2^{\ +} + C\ell O_4^{\ -}$$

The species that reacts with nicotine during the titration is $CH_3COOH_2^{\ +}$. 'Glacial' acetic acid does not contain any water.

The perchloric acid/acetic acid solution must be standardised before use and this can be done by titrating it with a solution made from a primary standard.

(a) Other than possessing a relatively high molar mass, state **two** characteristics required of a substance for it to be used as a primary standard. [2]

One: ____

____

Two: ____

____

A brand of nicotine patches comes in dose sizes of 7 mg, 14 mg and 21 mg. A manufacturing error produced a batch of unlabelled boxes of patches. A chemist was given the task of identifying the dose size so that the boxes could be accurately labelled and then sold.

The chemist took one of the boxes and extracted all the nicotine from the 14 patches it contained. The nicotine extract was then made up to a total of 100.0 mL using a suitable solvent. Aliquots of the resulting solution (20.0 mL) were then titrated with standardised 0.0483 mol L^{-1} perchloric acid/acetic acid solution, requiring an average of 15.11 mL to reach the end point.

(b) Complete the following table by writing the name of the most suitable piece of equipment to use for each task. [3]

Task	Piece of equipment to use
Making exactly 100.0 mL of nicotine-containing solution	
Measuring a 20.0 mL aliquot of the nicotine-containing solution	
Adding the perchloric acid/acetic acid solution to the nicotine-containing solution	

(c) Use the chemist's titration data to identify the nicotine dosage of the patches in the unlabelled boxes. Show all of your working.

The molecular formula of nicotine is $C_{10}H_{14}N_2$ and the titration reaction is:

$$C_{10}H_{14}N_2 + 2\ CH_3COOH_2^+ \rightarrow C_{10}H_{16}N_2^{2+} + 2\ CH_3COOH$$ [7]

9.(2020:27)

Iron filings and dilute hydrochloric acid

Equation $Fe(s) + 2H^+(aq) \rightarrow Fe^{2+}(aq) + H_2(g)$
Observation(s) A grey solid is added to a clear colourless liquid. A colourless odourless gas is formed and the solution becomes pale green.

Chromium(III) nitrate solution and magnesium ribbon

Equation $2Cr^{3+}(aq) + 3Mg(s) \rightarrow 3Mg^{2+}(aq) + 2Cr(s)$
Observation(s) A grey solid is added to a deep green solution. The grey solid dissolves, a grey solid forms and the deep green solution fades. This is an example of a metal displacement reaction because Mg is a more active reductant than Cr.

Potassium chloride solution and bromine water

Equation 'no reaction' – bromine cannot spontaneously displace chlorine.
Observation(s) 'no visible reaction' – an orange solution was added to a colourless solution and the mixture remained orange.

10.(2020:36) a) To be a functioning galvanic cell it must be spontaneous – the E° must be positive with 1 mol L^{-1} solutions. So magnesium and $Cu^{2+}(aq)$ meet this criterion. The cathode must not react with the $Cu^{2+}(aq)$ so it must be copper or graphite. The salt bridge should be ionic and soluble but not form precipitates so a carbonate would not be a good choice. (i) magnesium (ii) copper or graphite (iii) (1.0 mol L^{-1}) magnesium sulfate (solution) (iv) (1.0 mol L^{-1}) copper(II) sulfate (solution)

b) In the external wire an arrow from left to right: →

c) Oxidation occurs at the anode. Reduction occurs at the cathode.

Anode half-equation – $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$

Cathode half-equation – $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

d) Note the question asks for the units to be included

$Mg(s) \rightleftharpoons Mg^{2+}(aq) + 2e^-$ E° = +2.36V

$\underline{Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)}$ $\underline{E° = +0.34V}$

Total +2.70 V

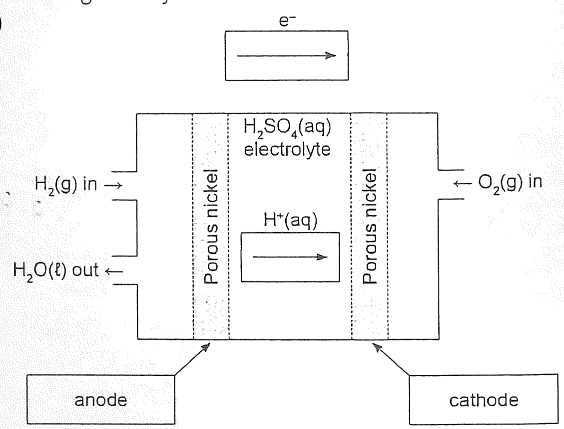
e) i) A salt bridge provides ions to each cell to maintain neutrality. Sometimes this is expressed as completing the circuit.

ii) In this cell, at the anode, positive $Mg^{2+}(aq)$ ions are being formed (oxidation) while at the cathode, $Cu^{2+}(aq)$ ions are being destroyed (reduction).

Negative ions move through the salt bridge to return the charge to zero at the anode half-cell where $Mg^{2+}(aq)$ ions are being formed.

Positive ions move through the salt bridge to return the charge to zero at the cathode half-cell where $Cu^{2+}(aq)$ ions are being destroyed and the solution would have been left with a negative charge.

11.(2021:30) a)



b) Oxidation reaction: $H_2(g) \rightarrow 2H^+(aq) + 2e^-$

- Reduction reaction: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow + 2H_2O(\ell)$
- Overall redox reaction: $2H_2(g) + O_2(g) \rightarrow 2H_2O(\ell)$
  - correct species
  - correct balancing

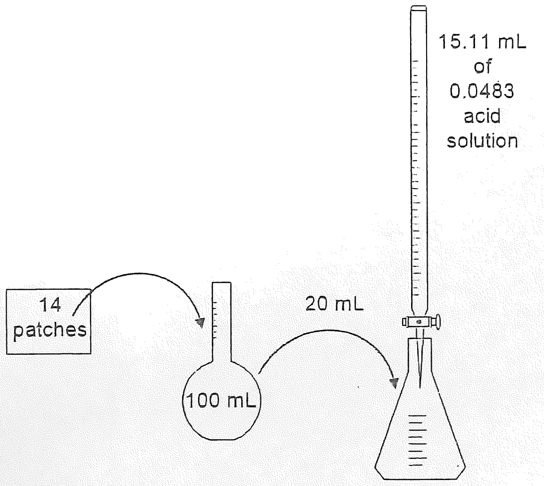
d) Phenolphthalein
From part (c)
Equation: $3\ NaOH + H_3PO_4 \rightarrow Na_3PO_4 + 3\ H_2O$
$\therefore$ PO_4^{3-} is the product of the reaction – it is the conjugate of the weak acid $HPO_4^{2-}(aq)$ and will undergo hydrolysis – $PO_4^{3-}(aq) + H_2O(\ell) \rightleftharpoons HPO_4^{2-}(aq) + OH^-(aq)$.
Thus the $[OH^-] > [H^+]$ and the pH >7
Of the three indicators, phenolphthalein changes colour (has an end point) at a suitable pH >7.

4.(2021:35) a) Any two relevant points. Answers could include:
- available in very high purity
- very low reactivity with CO_2 and/or O_2
- not hygroscopic
- (highly) soluble
- known purity
- not deliquescent
- predictable reactivity

b)

Making exactly 100.0 mL of nicotine-containing solution	volumetric flask
Measuring a 20.0 mL aliquot of the nicotine-containing solution	pipette
Adding the perchloric acid/acetic acid solution to the nicotine-containing solution	burette

c)



$C_{10}H_{14}N_2 + 2\ CH_3COOH_2^+ \rightarrow C_{10}H_{16}N_2^{2+} + 2\ CH_3COOH$
Burette:
n(acid) in 15.11 mL = c × V = 0.0483 × 0.01511 = 7.298×10^{-4} mol
Flask:
n(nicotine) in 20.0 mL = 0.5 × 0.0007298 = 0.0003649 mol
100 mL Volumetric Flask
n(nicotine) in 100.0 mL = 0.0003649 × 5.000 = 0.0018245 mol
Molar mass nicotine
M(nicotine) = (10 × 12.01) + (14 × 1.008) + (2 × 14.01) = 162.232 g mol^{-1}
Box:
m(nicotine) in 100.0 mL = 162.232 × 0.0018245 = 0.29599 g
m(nicotine) in 1 patch = 0.29599/14 = 0.02114 g
The boxes contain 21 mg patches

Chapter 14A: Calculations: Organic

1.(2016:38) a) Enzymes are proteins which act as biological catalysts. They are more efficient than inorganic catalysts.
b)

Ethanoic acid	Choline